3.3 Django shell

As templates are typically rendered through a view, learning and experimenting with them might mean having to write a lot of throwaway code. Instead of creating a dummy view, you can use the Python read, evaluate, print loop (REPL). If you haven't come across the REPL yet, it's built into Python and allows you to immediately see the results of a line of code. In the REPL, you type some code (the tool *reads* it), and then the tool *evaluates* it, *prints* the results, and *loops* to do it all over again.

For regular Python, you enter the REPL by running `python` without any arguments. With Django being a framework, you can't just fire up the Python REPL within a Django context. There are several steps to be performed to get you into "Django land," including importing and activating the framework. Luckily, Django provides a command, so you don't have to do this by hand. Instead of calling `python`, you call the `manage.py` command named `shell`:

```
(venv) RiffMates$ python manage.py shell
```

When you run the command, you get a regular Python REPL, except it is Django aware. Because Django is a framework, if you used the default Python REPL on its own, you'd have to run some commands to get it to load and understand the Django environment; instead, the `shell` command takes care of this for you. When started, the command reports version information and then sends you to the `>>>` prompt, awaiting your instructions:

```
(venv) RiffMates$ python manage.py shell
Python 3.12.1 [Clang 13.0.0 (clang-1300.0.29.30)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
(InteractiveConsole)
>>>
```

If you need to experiment with some Django code, using the `shell` command to enter the REPL is less work than trying to implement your idea in a view. It is also less error prone; you don't have to remember to remove your experiment afterward. Now, let's use the `shell` to play with some templates.

> **NOTE** The REPL supports the use of the up arrow to retrieve previous commands. This can save a lot of typing if you're doing repetitive work. If you are using the REPL a lot, you may want to search for third party replacements, such as `bpython`, `IPython`, and `ptpython`, that include advanced code editing tools and macro capabilities.

3.4 Rendering a Template

`Template` objects are, well, objects. You can instantiate them like any other object. In practice, you're more likely to use a shortcut that loads a file containing the template content, but let's deal with the objects directly first to better understand how they work. You'll learn about using shortcuts to load and render template content from file a little later in the chapter.